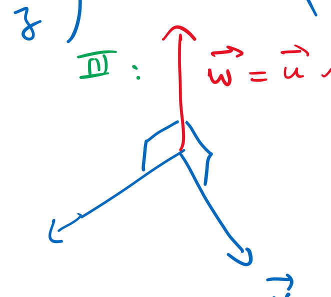
Annexe : produit vectoriel ↳ cross product] Outil de $\mathbb{R}^3$.

$\vec{u} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\vec{v} = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$



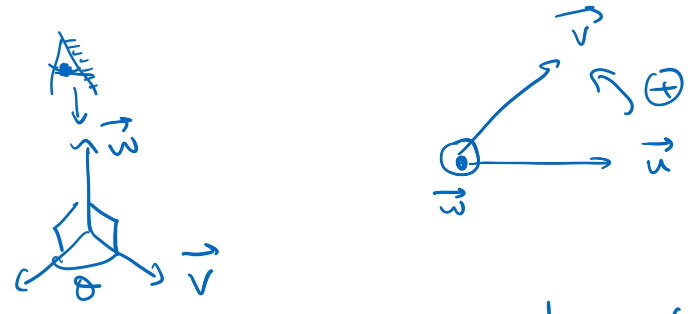
produit vectoriel

$$\vec{w} = \vec{u} \wedge \vec{v}$$

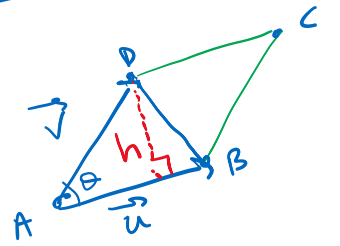
$$\underbrace{\underset{\text{majeur}}{\text{III}} \quad \underset{\text{pouce}}{\text{I}} \quad \underset{\text{index}}{\text{II}}}_{\text{main droite.}}$$

On a 1) $\vec{w} \perp \vec{u}$ et $\vec{w} \perp \vec{v}$.

2) $(\vec{u}, \vec{v}, \vec{w})$: sens direct.



ABCD : parallélogramme.



θ : aigu

3) $\|\vec{u} \wedge \vec{v}\| = \|\vec{u}\| . \|\vec{v}\| . |\sin(\theta)|$.

$\mathcal{A}_{ABCD} = 2 \mathcal{A}_{ABD}$

$= \cancel{2} \times \dfrac{AB \times h}{\cancel{2}}$

$\sin\theta = \dfrac{h}{AD} \Leftrightarrow h = \sin\theta \, AD$

$\Rightarrow \mathcal{A}_{ABCD} = \underbrace{AB}_{\|\vec{u}\|} . \underbrace{AD}_{\|\vec{v}\|} \underbrace{\sin\theta}_{>0,\text{ car } \theta \text{ aigue}}$.

Formule

$$\begin{array}{ccc} \vec{u} & & \vec{v} \\ \begin{pmatrix} x \\ y \\ z \end{pmatrix} & \wedge & \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \\ x & & x' \\ y & & y' \end{array} = \overset{\vec{w}}{\begin{pmatrix} yz' - zy' \\ zx' - xz' \\ xy' - yx' \end{pmatrix}}$$

Ex 1) Calculer $\vec{u} \wedge \vec{v}$ avec $\vec{u} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix}$

$$\begin{array}{ccc} \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} & \wedge & \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix} \\ 1 & & 4 \\ 2 & & -2 \end{array} = \begin{pmatrix} -2-6 \\ -12+1 \\ -2-8 \end{pmatrix} = \begin{pmatrix} -8 \\ -11 \\ -10 \end{pmatrix} = \vec{u} \wedge \vec{v}.$$

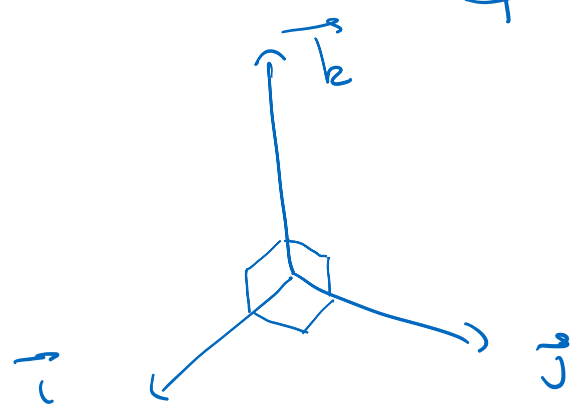
2) A quelle condition $\vec{u} \wedge \vec{v} = \vec{0}$

$\|\vec{u} \wedge \vec{v}\| = \|\vec{u}\| . \|\vec{v}\| \, |\sin\theta| = 0 \Leftrightarrow \left| \begin{array}{l} \vec{u} = \vec{0} \text{ ou } \vec{v} = \vec{0} \\ \text{ou } \left| \begin{array}{l} \theta = 0 \, [\pi] \\ \theta = k\pi, \, k \in \mathbb{Z} \end{array} \right. \end{array} \right.$

$\theta = (\vec{u}, \vec{v})$

rq si $\vec{u}$ et $\vec{v}$ sont colinéaires, $\vec{u} \wedge \vec{v} = \vec{0}$

3)



donner

$\vec{i} \wedge \vec{j} = \vec{k}$ | $\vec{k} \wedge \vec{j} = -\vec{i}$
$\vec{j} \wedge \vec{i} = -\vec{k}$ | $\vec{i} \wedge \vec{k} = -\vec{j}$.
$\vec{k} \wedge \vec{i} = \vec{j}$
$\vec{j} \wedge \vec{k} = \vec{i}$

prop

1) $\vec{u} \wedge \vec{v} = -\vec{v} \wedge \vec{u}$

2) $\alpha(\vec{u} \wedge \vec{v}) = (\alpha\vec{u}) \wedge \vec{v} = \vec{u} \wedge (\alpha\vec{v})$
$\alpha \in \mathbb{R}$

3) $\vec{u} \wedge (\alpha\vec{v} + \beta\vec{w}) = \alpha(\vec{u} \wedge \vec{v}) + \beta(\vec{u} \wedge \vec{w})$
$\alpha, \beta \in \mathbb{R}$

4) $\wedge$: n'est pas associatif, càd
$\vec{u} \wedge (\vec{v} \wedge \vec{w}) \neq (\vec{u} \wedge \vec{v}) \wedge \vec{w}$.